Chapter 1

Su-Schrieffer-Heeger model with multiorbital superconductivity

As a first departure into the search for systems exhibiting topological superconductivity, we begin with a topologically non-trivial system. A canonical topological system is the Su-Schrieffer-Heeger (SSH) model of a spinless, one-dimensional dimer lattice of fermions with staggered hopping. The model obeys a *chiral symmetry*, exhibiting two topological phases. In the topologically non-trivial phase, a pair of edge states emerges at the boundaries, characterised by a *winding number*. We will consider the effect of a spin-triplet, multiorbital attractive interaction within the unit cell, and a Coulomb repulsion to nearest-neighbours outside of the unit cell.

1.1 Su-Schrieffer-Heeger model with spin

The Su-Schrieffer-Heeger model with additional spin degrees of freedom is given by the following Hamiltonian

$$\hat{\mathcal{H}}_{\text{SSH}} = \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow\downarrow\}} \sum_{m\in\{A,B\}} \left[-\mu \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m}\right] + \sum_{\sigma\in\{\uparrow\downarrow\}} \left[V \hat{c}^\dagger_{0,\sigma,m} \hat{c}_{0,\sigma,m}\right] + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow\downarrow\}} \left[-v \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} - w \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} + \text{h.c.}\right], \quad (1.1)$$

where μ is the chemical potential, v is the intracell, interorbital hopping parameter, w is the intercell, interorbital hopping and V is an impurity coupling strength. The model boundary conditions are periodic, and therefore, in the topological state, we expect edge modes to arise around the impurity. In the trivial state, with $v > w$, the unit cell has the strucure displayed in Figure 1.1.

We investigate the phases of the model in Figure 1.2. Importantly, we may have naivly expected the addition of a spin quantum number to the SSH to be trivial i.e. simply doubling the degrees of freedom of the original SSH model. Performing the calculation, however, reveals nontrivial magnetic effects at $\mu = 0$.

1.2 Non-unitary spin-triplet Su-Schrieffer-Heeger model

The non-unitary spin-triplet extension of the Su-Schrieffer-Heeger model is given by

$$\hat{\mathcal{H}}_{\text{SSH-triplet}} = \hat{\mathcal{H}}_{\text{SSH}} + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[-U_v \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} + U_w \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \right], \tag{1.2}$$

where U_v is the spin-triplet interaction and U_w is a Coulomb repulsion.

The mean-field solution is

$$\begin{aligned} \hat{\mathcal{H}}^{\text{mf}}_{\text{SSH-triplet}} = \hat{\mathcal{H}}_{\text{SSH}} + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \Bigg[& \sum_{m\in\{A,B\}} \left[\phi_{x,\sigma,m} \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m} \right] \\ & + \Phi^v_{x,\sigma} \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} + \Phi^w_{x,\sigma} \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} + \Delta^v_{x,\sigma} \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} + \Delta^{v\dagger}_{x,\sigma} \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} \\ & + \Delta^w_{x,\sigma} \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} + \Delta^{w\dagger}_{x,\sigma} \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \Bigg], \end{aligned} \tag{1.3}$$

where the Hartree, Fock and Gorkov fields are defined as

$$\phi_{x,\sigma,m} = (U_w - U_v) \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m} \tag{1.4}$$

$$\Phi^v_{x,\sigma} = -U_v \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} \tag{1.5}$$

$$\Phi^w_{x,\sigma} = \hat{U}_w c^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} \tag{1.6}$$

$$\Delta^v_{x,\sigma} = -U_v \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} \tag{1.7}$$

$$\Delta^{v\dagger}_{x,\sigma} = U_w \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} \tag{1.8}$$

$$\Delta^v_{x,\sigma} = -U_v \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \tag{1.9}$$

$$\Delta^{w\dagger}_{x,\sigma} = U_w \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} \tag{1.10}$$

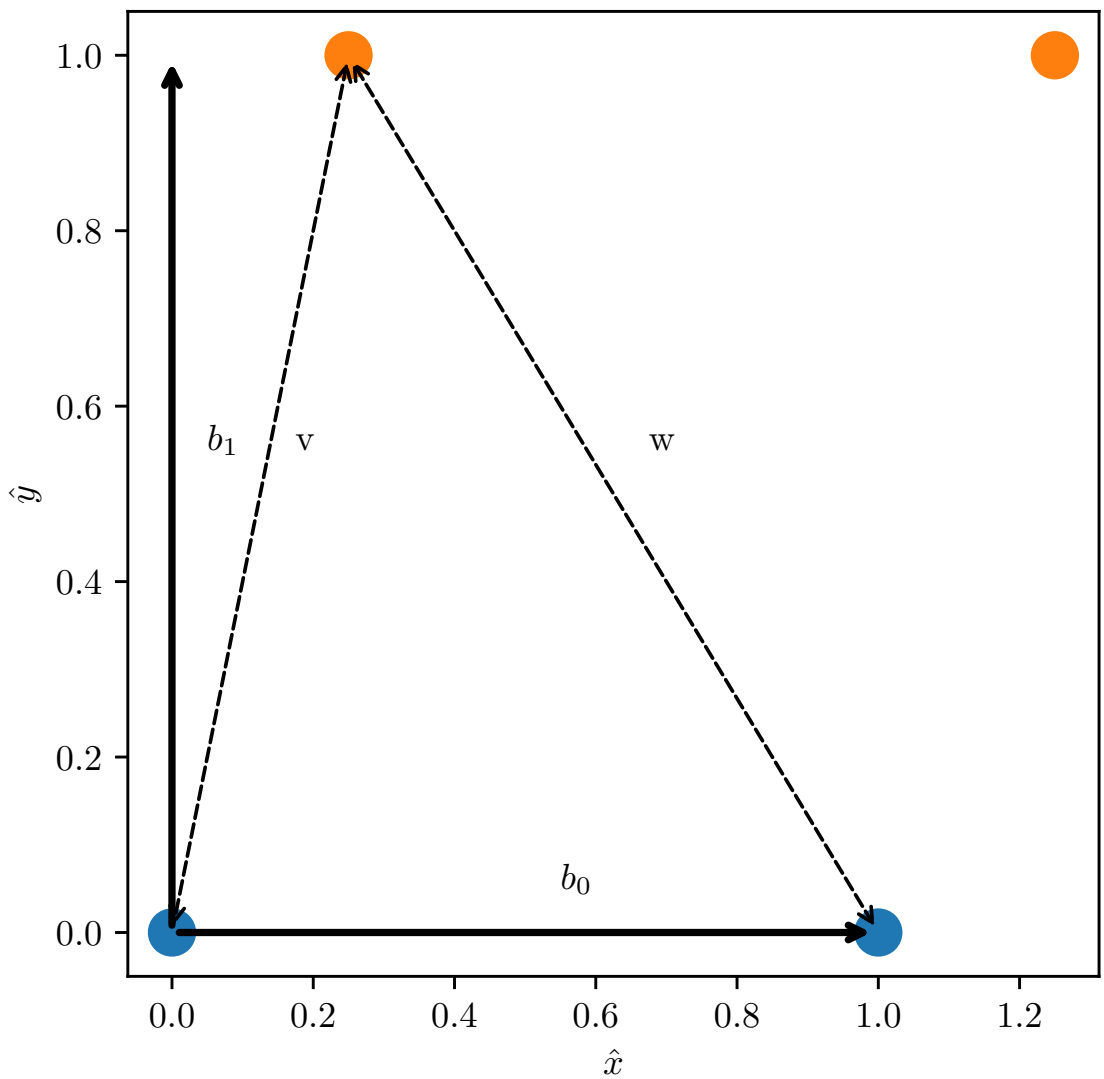

Figure 1.1: The dimer unit cell of the SSH model is depicted, together with hopping parameters w and v, as well as the lattice basis b_0 and b_1. Atoms A and B are colored in blue and orange, respectively.

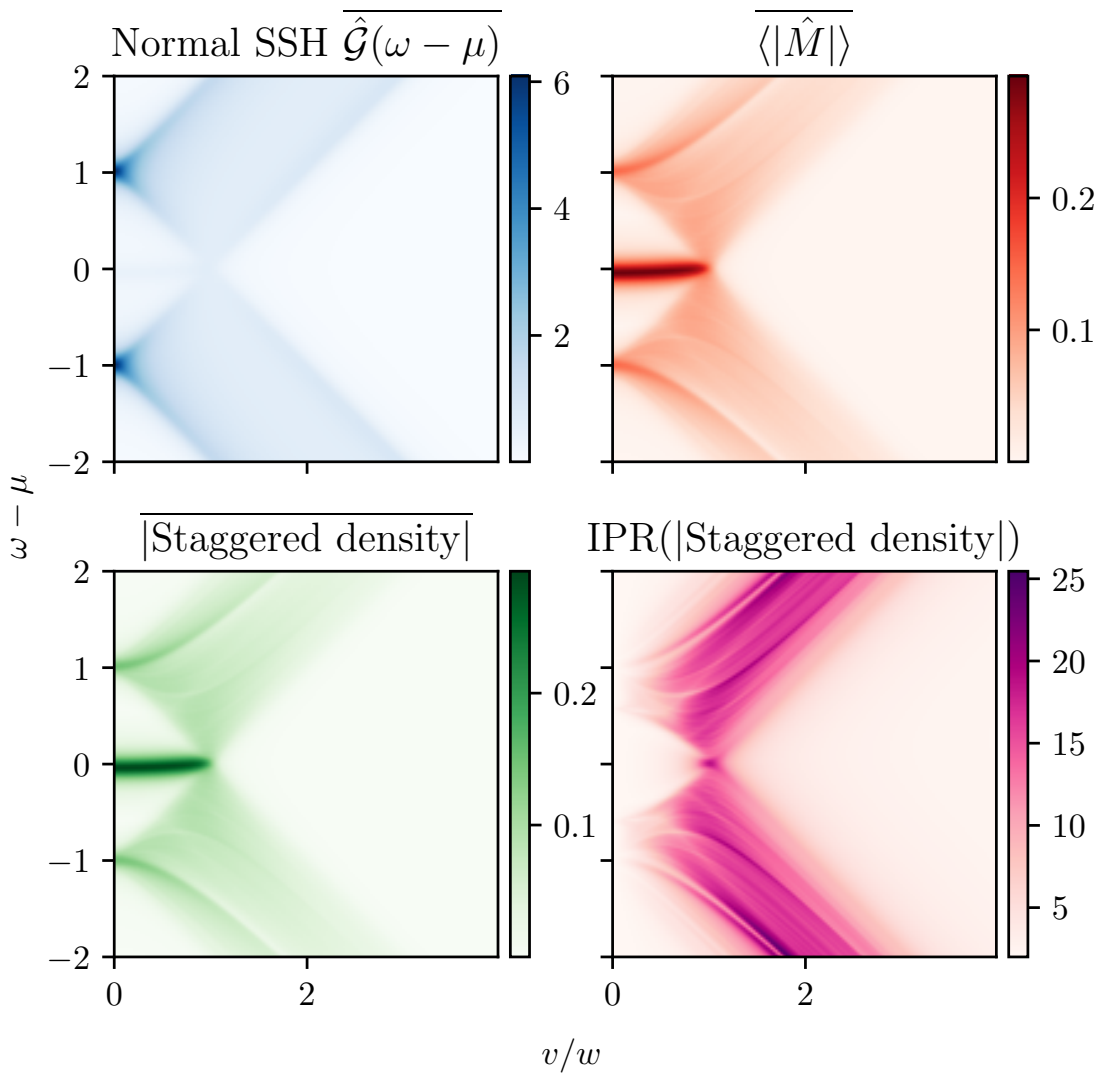

Figure 1.2: Phase diagram of the SSH model with $n = 43$ dimers and periodic boundary conditions. The density of states are plotted as a function of the applied bias energy $\omega - \mu$, noting that the chemical potential contributes only an overall shift, and as a function of the intracell, interorbital hopping parameter v/w. The topological transition occurs at $v/w = 1$ and notice the faint density at $\omega = 0$ below the transition point, due to the edge states. A large impurity coupling $V = 20.28$ gaps out any modes on the middle site, effectively cutting the closed chain into open boundary conditions. In order to have a measure of the topological modes, we plot the mean absolute magnetism and the mean absolute staggered density, which signal charge density waves. The inverse participation of the staggered density is a measure of long-range order, demonstrating that the topological modes are localised around the impurity.

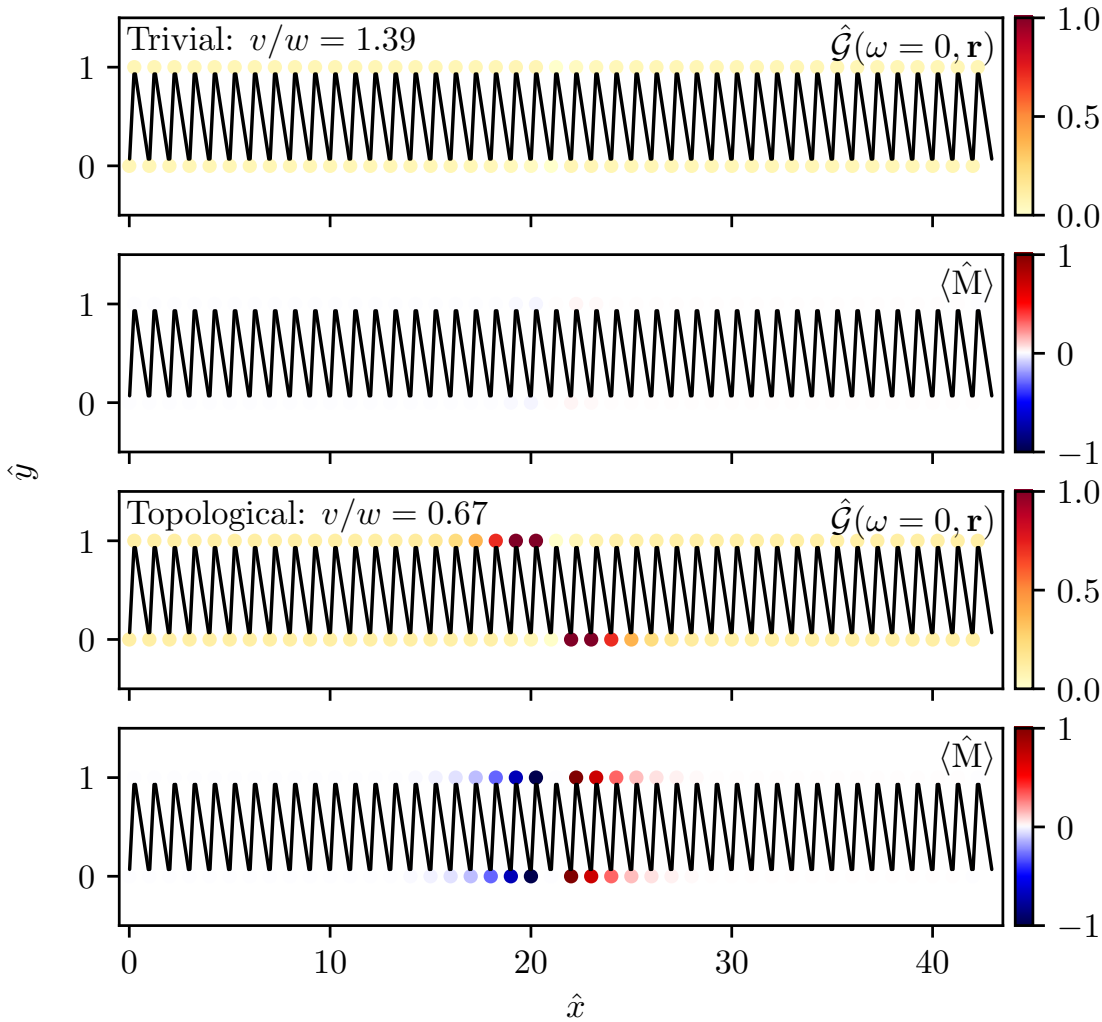

Figure 1.3: The local density of states and magnetism of the SSH model in the normal state with chemical potential $\mu = 0$ and strong impurity coupling strength $V = 20.28$ on the middle site, in order to cut the chain without weak coupling effects. The dark lines represent hopping bonds and the sites are denoted by circles with color corresponding to density. Importantly, we note non-trivial orbital and magnetic characteristics to the edge modes.

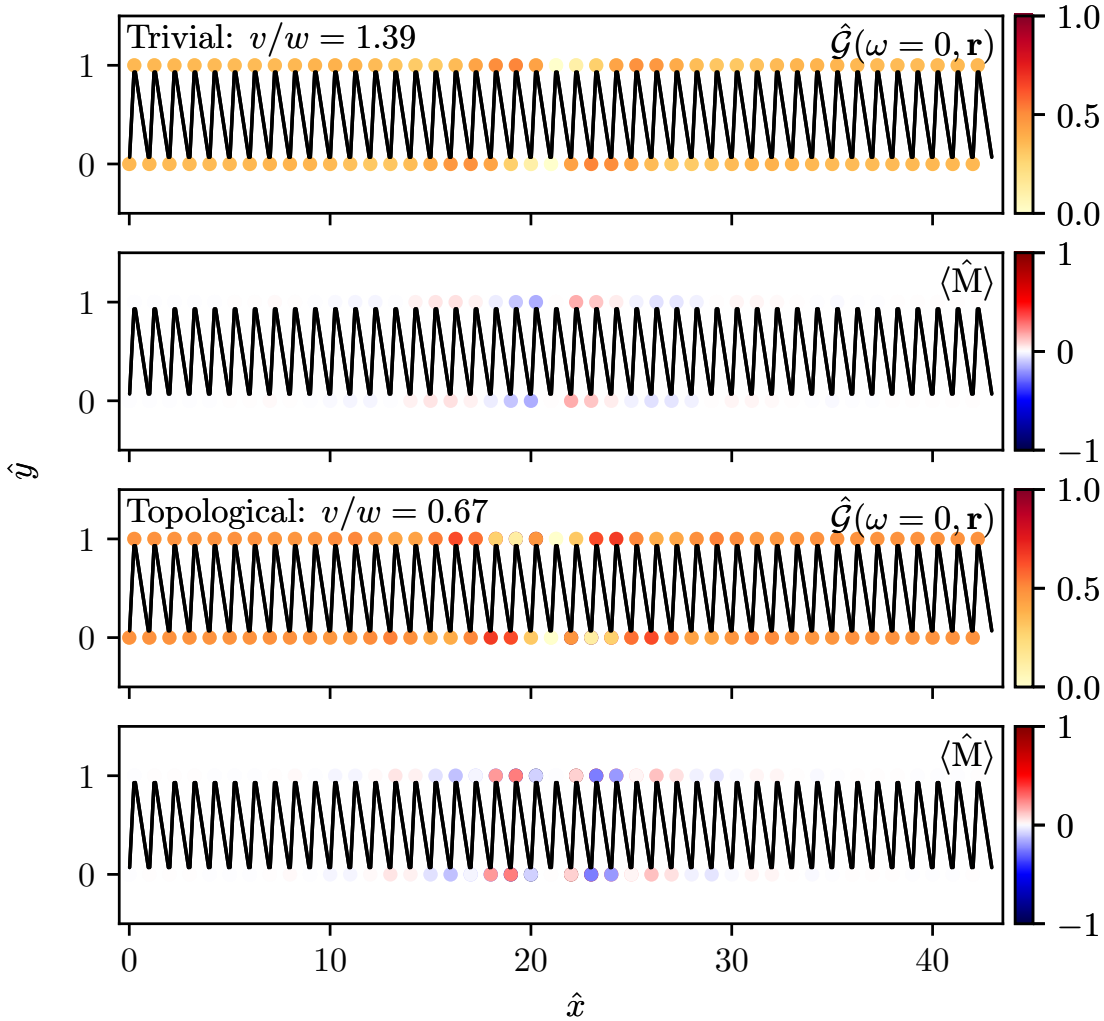

Figure 1.4: The local density of states and magnetism of the SSH model in the normal state with chemical potential $\mu = -0.57$ and strong impurity coupling strength $V = 20.28$ on the middle site. Away from zero chemical potential, we observe a charge density wave in the topological phase.

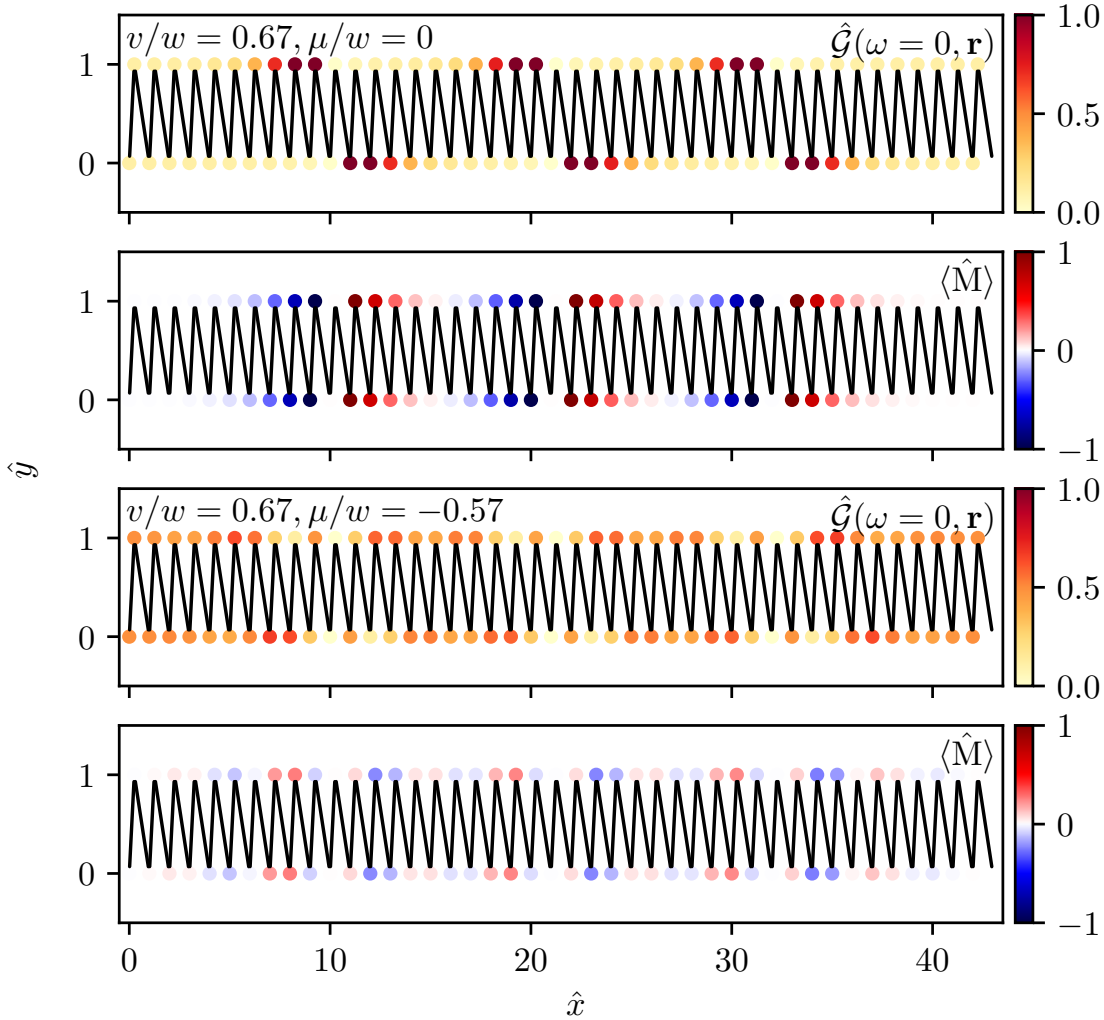

Figure 1.5: The local density of states and magnetism of the SSH model in the normal state with two impurites with strong coupling strengths $V = 20.28$ at $x = \pm 11$. Curiously, a kink forms in the middle $x = 0$, where there is no impurity.

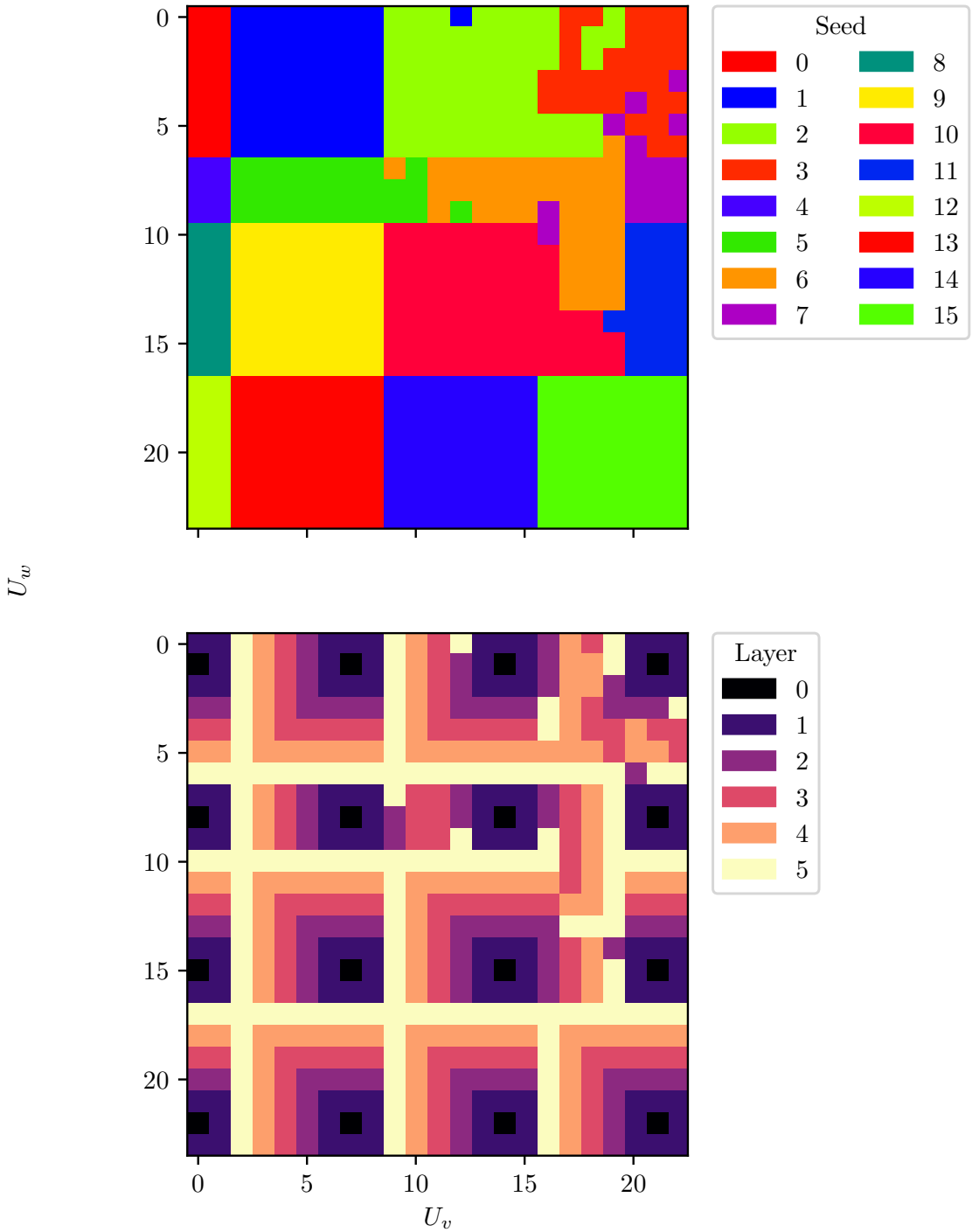

Figure 1.6: Building the phase diagram for a spin-rtiplet SSH model. Above: seeds are perculated evenly spaced throughout phase space. At each location, two seeds are placed, corresponding to the unitary and non-unitary initial mean-fields. Below: from these seeds (refered to as Layer 0), layers are grown outwards from their parent seed. The outer layers of various seeds overlap, and that with the minimum free energy is selected.

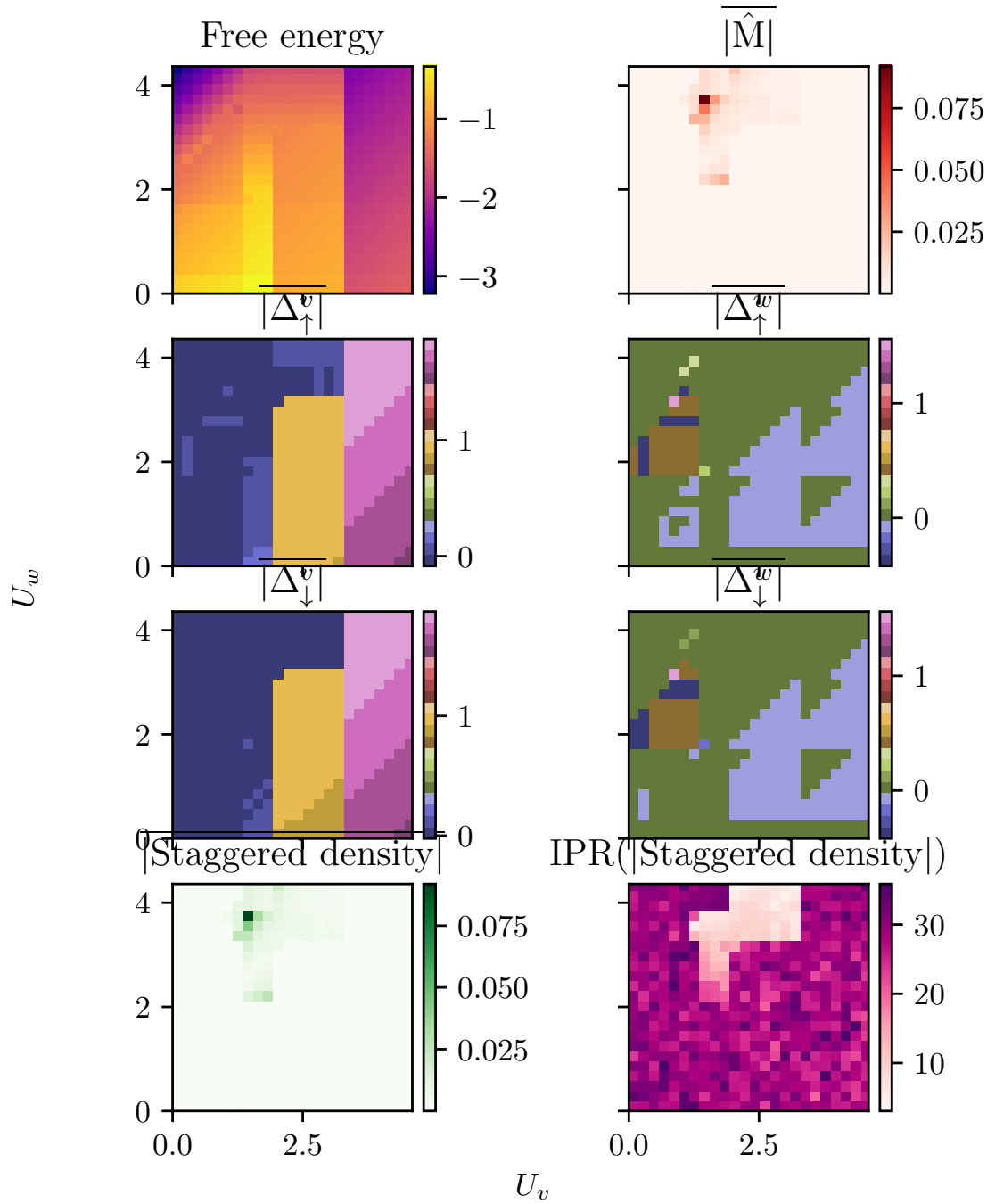

Figure 1.7: The phase diagrams for an SSH model with multiorbital, equal-spin superconductivity interaction and no impurity. Two initial pairing fields are given to the self-consistent loop: non-unitary triplet and equal-spin triplet. The solution with the lowest free energy is selected. We observe that the mean-field spin-sectors are equal $\overline{\Delta^{\uparrow\uparrow}_{v/w}} = \overline{\Delta^{\downarrow\downarrow}_{v/w}}$, hence the equal-spin triplet theory is favoured. We also observe spontaneous formation of charge density waves, localised in space.

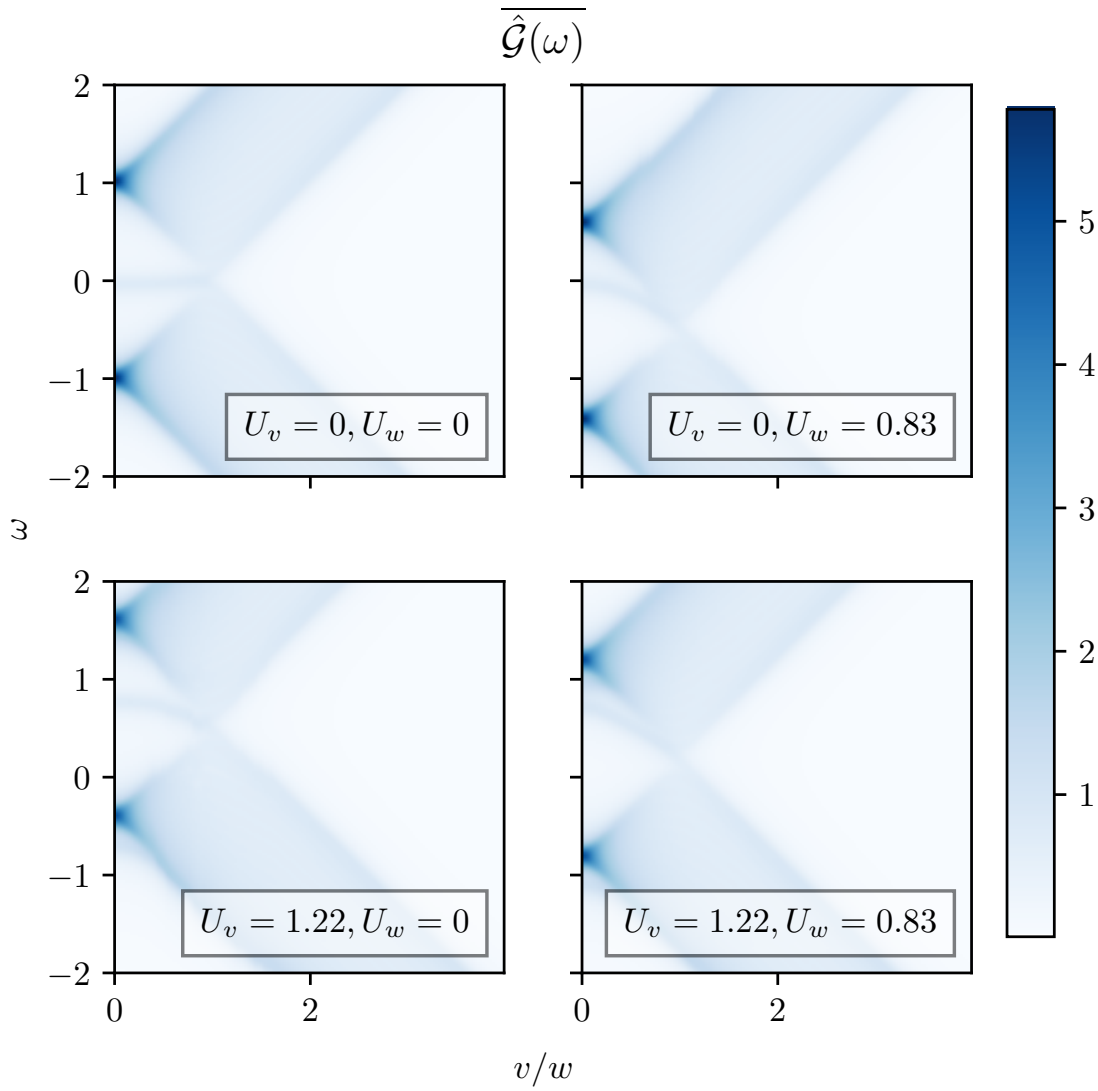

Figure 1.8: Mean value of the local density of states. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

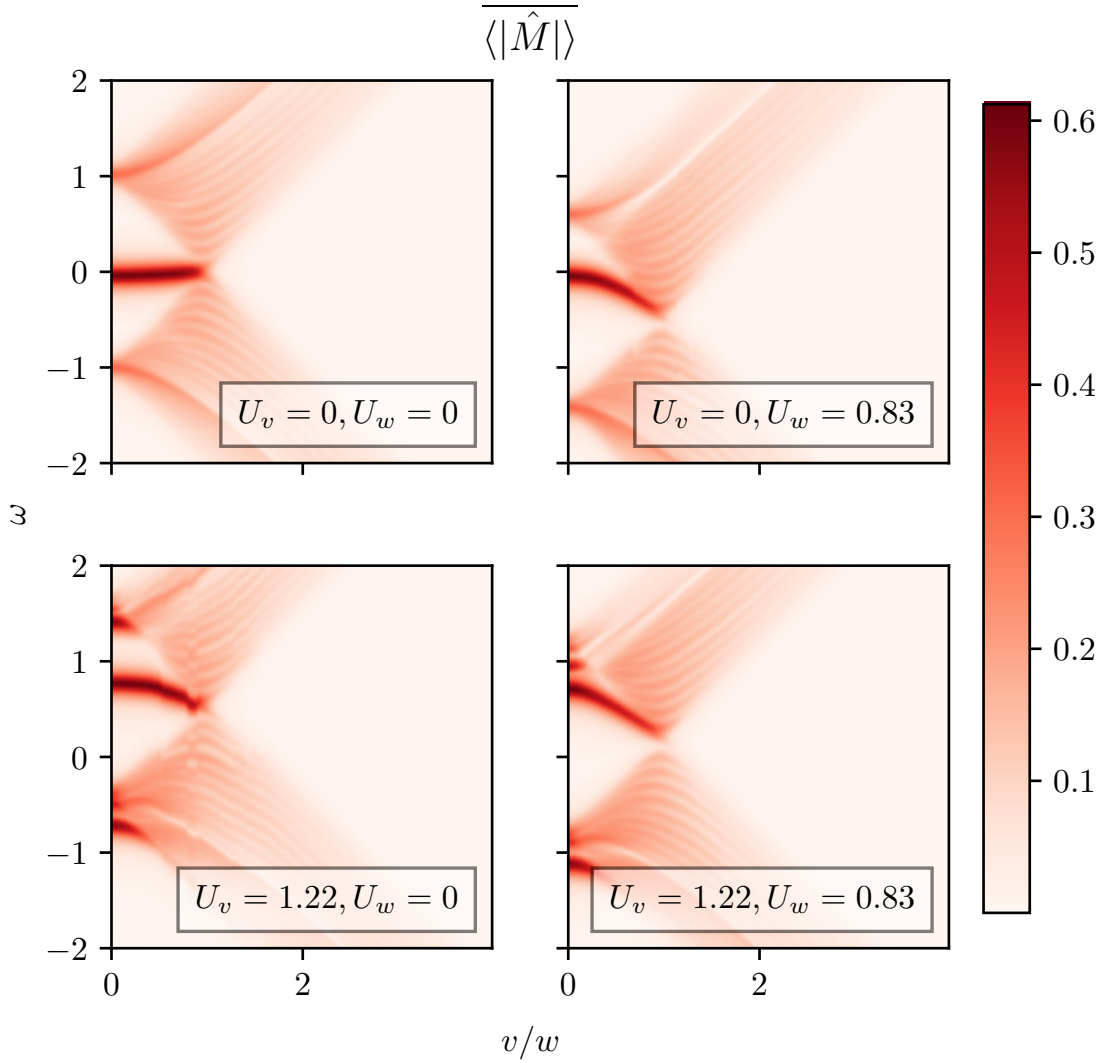

Figure 1.9: Mean absolute value of magnetism. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

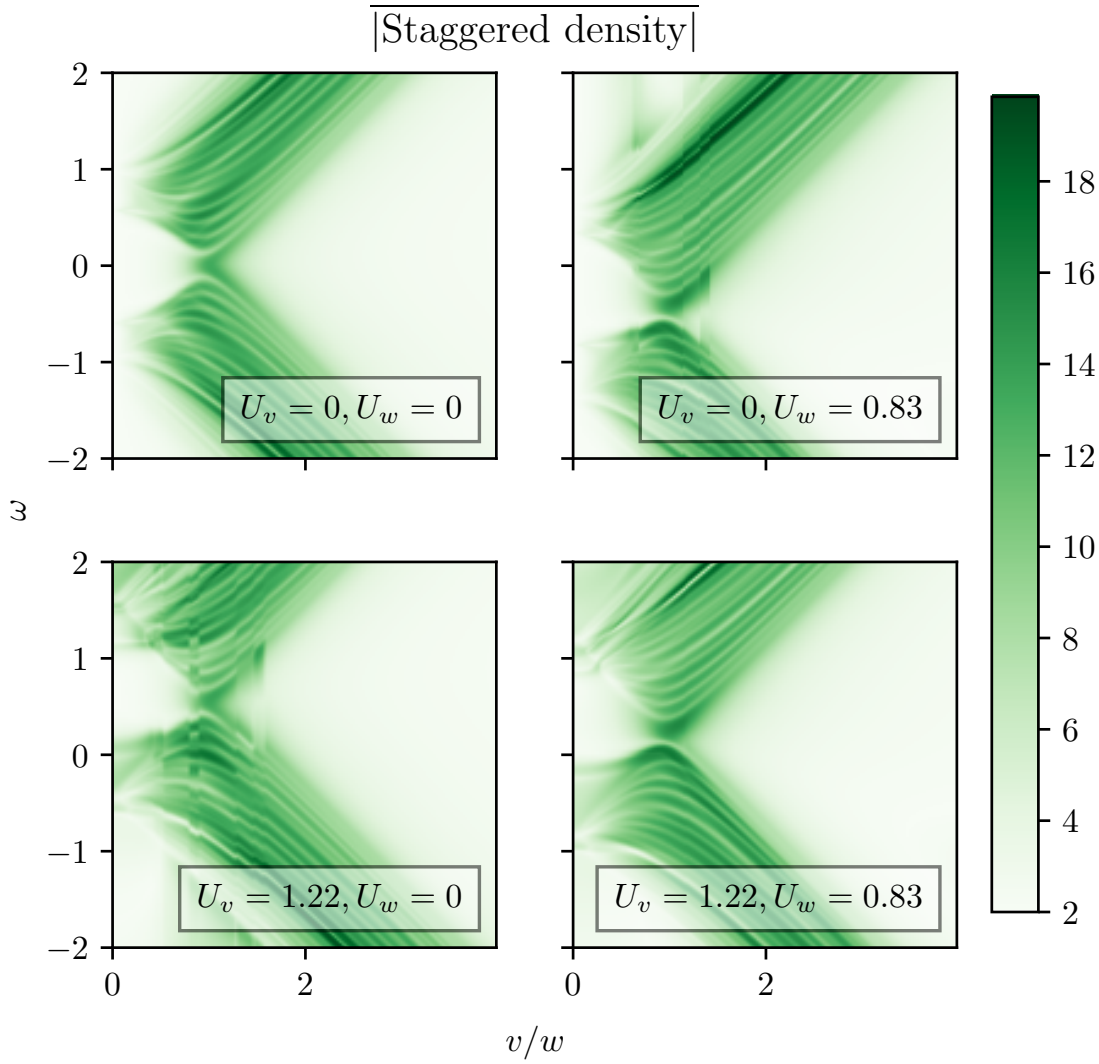

Figure 1.10: Mean staggered density. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

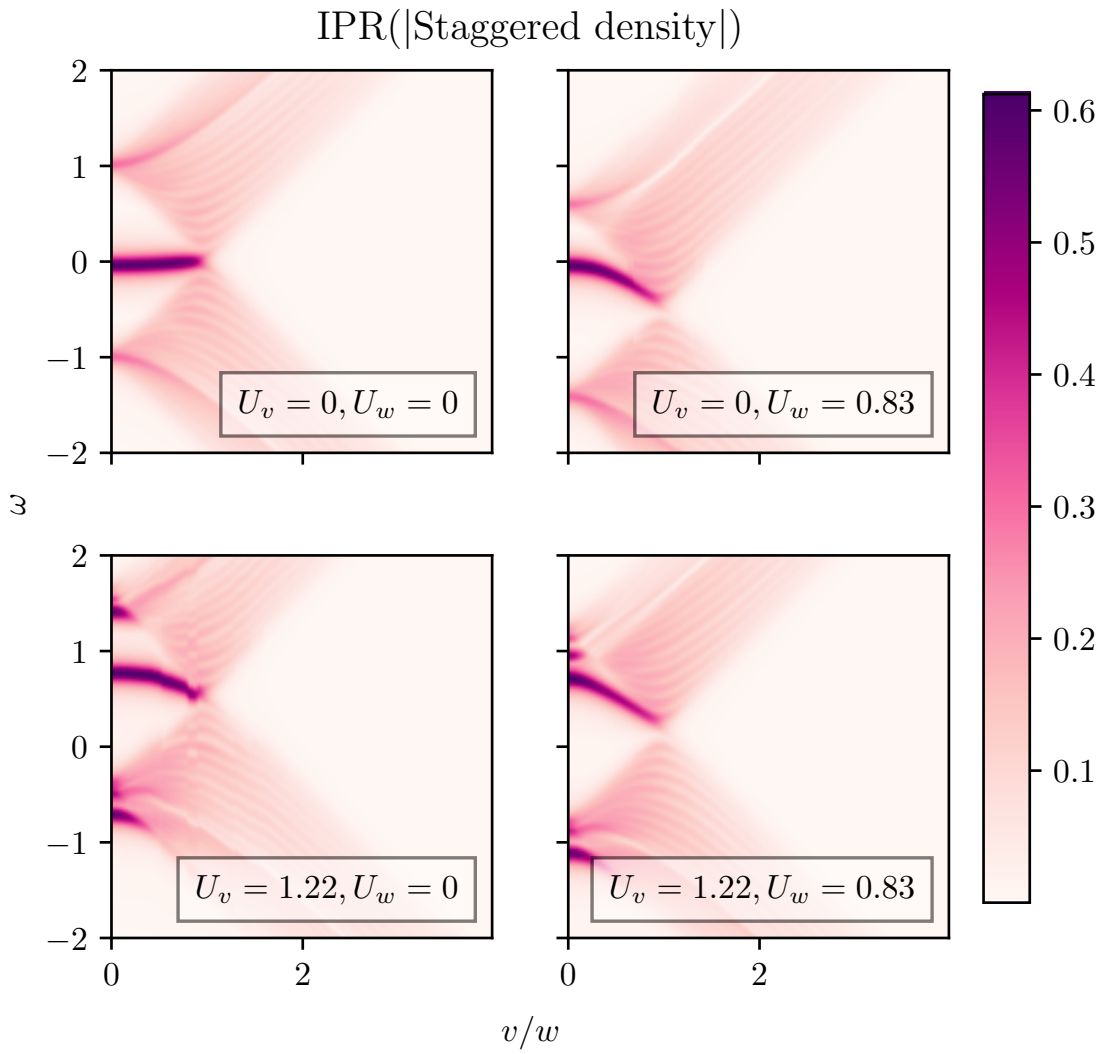

Figure 1.11: Inverse participation ratio of the staggered density, as a measure of correlation distance of the charge density wave. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.